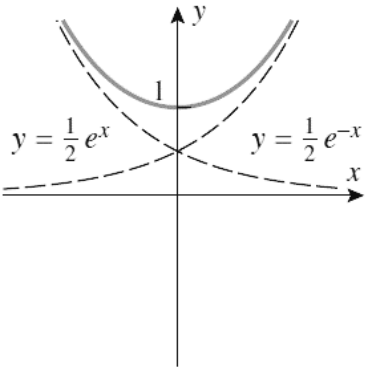
SOLUÇÕES FICHA 4

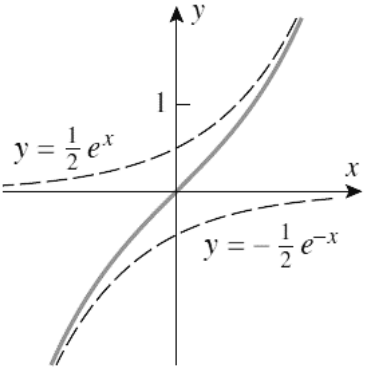
C. TÓPICOS ADICIONAIS

C.1. Funções hiperbólicas

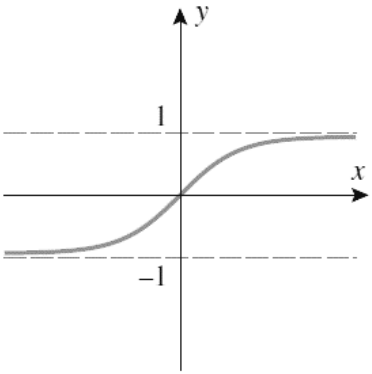
4.1.



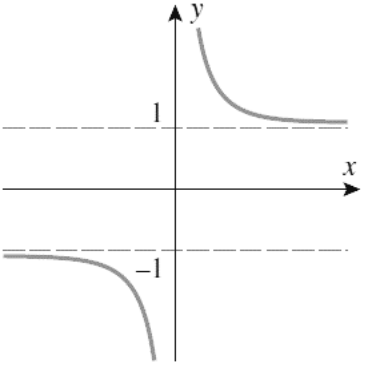
$\cosh x$
$D = (-\infty, +\infty);\ D' = [1, +\infty)$



$\sinh x$
$D = (-\infty, +\infty);\ D' = (-\infty, +\infty)$



$\tanh x$
$D = (-\infty, +\infty);\ D' = (-1,1)$



$\coth x$
$D = (-\infty, 0) \cup (0, +\infty)$
$D' = (-\infty, -1) \cup (1, +\infty)$

4.2.

a) $f(x) = \cosh(x) + 2\sinh(x) + \operatorname{sech}(x)\tanh(x)$

b) $f(x) = \sec^2(x) + 2x\,\operatorname{csch}(x) - x^2\coth(x)\operatorname{csch}(x)$

c) $f(x) = \dfrac{\tanh(x) - (x+1)\operatorname{sech}^2(x)}{\tanh^2(x)}$

d) $f(x) = 3x\sinh(x^3)$

e) $f(x) = \operatorname{sech}^2(x)\coth(x)$

4.3.

a) $\ln[\cosh(x)] + C$

b) $\ln|\sinh(x)| + C$

c) $2\arctan(e^x) + C$

d) $\begin{cases} -2\operatorname{arctanh}(e^x) + C, & e^x < 1 \\ -2\operatorname{arccoth}(e^x) + C, & e^x > 1 \end{cases}$

e) $\sinh(x) + \dfrac{2}{3}\sinh^3(x) + \dfrac{1}{5}\sinh^5(x) + C$

f) $-\dfrac{1}{8}x + \dfrac{1}{36}\sinh(4x) + C$

g) $\dfrac{1}{a}\ln[\cosh(ax)] + C$

h) $-\dfrac{1}{a\cosh(ax)} + C$

i) $\dfrac{1}{2}e^{\cosh(2x)} + C$

j) $2\cosh\left(\sqrt{x}\right) + C$

k) $\ln[1 + \cosh(x)] + C$

l) $-\dfrac{1}{3}\operatorname{sech}^3(x) + C$

m) $\dfrac{1}{2}\tanh(x^2) + C$

n) $\dfrac{1}{2}\{\ln[\cosh(x)]\}^2 + C$

o) $\tanh(x) - \dfrac{1}{2}\operatorname{sech}^2(x) + C$

p) $\ln|1 + \tanh(x)| + C$

r) $\dfrac{1}{6}\tanh^6(x) + C$

C.2. Integrais impróprios

4.4.

a) 2 **b)** $\frac{\ln(10)}{4[\log_{10}(2)]^4}$ **c)** 1

d) $\frac{\pi}{2}$ **e)** $+\infty$ (divergente) **f)** $+\infty$ (divergente)

4.5.

a) $\frac{3}{2}$ **b)** $+\infty$ (divergente) **c)** $-\frac{1}{\ln(10)}$

d) $+\infty$ (divergente) **e)** 2 **f)** π

4.6.

$a = 2e - 2$

$b = 2e - 2$

4.7.

a) --- **b)** ---

4.8.

a) Divergente **b)** Divergente

C.3. Equações diferenciais Ordinárias de 1.ª e 2.ª ordem

4.9.

a) $y^2 = 2x^2 + K$ ou $y^2 = 2(x^2 + c)$

b) $y^2 = \frac{4}{9}\sqrt[2]{x^3} + K$ ou $y^2 = 2\left(\frac{2x^{3/2}}{9} + c\right)$

c) $y = \frac{x^2}{2} + 2x + c$

d) $y = c(1 + x^2)$

e) $y = -ce^{-\frac{x^2}{2}} + 100$

4.10.

a) $y = \frac{(\ln|x|)^2}{2} + 2$

b) $y^2 = 4(x^2 + 1) - 1$

c) $u = e^{\frac{1}{2} - \frac{\cos(v^2)}{2}}$

d) $T = 70e^{-kt} + 70$

4.11.

a) $y = \frac{e^x}{2} + \frac{c}{e^x}$

b) $y = x^2 \ln|x| + cx^2$

c) $y = \tan(t) + c\sec(t)$

d) $y = \frac{3x^2}{4} + \frac{2x}{3} + \frac{c}{x^2}$

4.12.

a) $y = 1 + 4e^{-\tan(x)}$

b) $y = -\frac{1}{2x^2}e^{\left(\frac{1}{x^2}\right)} + \frac{3}{2}e^{\left(\frac{1}{x^2}\right)}$

c) $y = \sin(x) + [x + 1]\cos(x)$

d) $y = 1 + \frac{3}{\sec(x) + \tan(x)}$

e) $y = x(\ln|x| + 12) - 2$

4.13.

a) $y^2 = \dfrac{3}{1+ke^{2x^3}}$

b) $y^2 = 1 + ce^{-x^2}$

c) $y = \dfrac{1}{-x^2+cx}$

d) $y^2 = 8e^{3x} + 12ce^{\left(\frac{8}{3}x\right)} + 6c^2e^{\left(\frac{7}{3}x\right)} + c^3e^{(2x)}$

e) $y^2 = ce^{4x} - \dfrac{2}{3}e^x$

4.14.

a) $y = C_1 + C_2e^{-x}$

b) $y = C_1e^{-4x} + C_2xe^{-4x}$

c) $y = C_1\cos(3x) + C_2\sin(3x)$

d) $y = C_1e^{2x} + C_2e^{-2x}$

e) $y = C_1e^{-x} + C_2xe^{-x}$

f) $y = C_1e^{3x} + C_2e^{2x}$

g) $y = [C_1\cos(6x) + C_2\sin(6x)]e^{-2x}$

4.15.

a) $y = C_1e^{3x} + C_2e^{-2x} - 3x^2 + x - 2$

b) $y = C_1e^{-2x} + C_2e^{-5x} + \dfrac{1}{40}e^{3x}$

c) $y = C_1e^{-2x} + C_2e^{-5x} + \dfrac{x}{3}e^{-2x}$

d) $y = [C_1\cos(4x) + C_2\sin(4x)]e^{-3x} + \dfrac{1}{73}\cos(4x) + \dfrac{8}{219}\sin(4x)$

e) $y = C_1\cos(4x) + C_2\sin(4x) + \dfrac{1}{8}x\cos(4x)$

f) $y = K_1e^{3x} + K_2xe^{3x} + \dfrac{x^2}{2}\ln(x)\,e^{3x} - \dfrac{3}{4}x^2e^{3x}$

g) $y = K_1\cos(3x) + K_2\sin(3x) + \dfrac{\cos(3x)}{9}\ln|\cos(3x)| + \dfrac{x\sin(3x)}{3}$